
GNN-Based BERT for Understanding Context from Music

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Abstract

Music carries genre, mood, structure, and free-form captions at the same time. Spectrogram CNNs model local sound well, but they do not clearly represent how segments relate over time. In this CSE425 project we build a DistilBERT + GraphSAGE system for four tasks: multi-label tagging from MusicCaps captions, FMA-small genre classification from segment graphs, cross-attention fusion of graph and text with DEAM emotion auxiliaries, and InfoNCE caption–audio retrieval. On our real-data run (2,000 FMA tracks, 800 MusicCaps audio graphs, DEAM), BERT tagging reaches Macro-F1 0.71 and AUC-PR 0.78; GraphSAGE matches a mel-CNN baseline at about 0.30 Macro-F1; fusion improves over GNN-only; and retrieval reaches $R@5 = 0.34$. We include baselines, ablations, plots, qualitative examples, and a runnable repository.

1 Introduction

A short clip can be electronic, melancholic, and described by a detailed caption all at once. “Context” therefore mixes tags, affect, and language [2, 3, 6]. CNNs on log-mel spectrograms are strong local detectors, but they flatten how neighboring windows evolve and how similar segments form a song graph.

This project follows the CSE425 brief [1]: combine a BERT-family text encoder with a graph neural network for understanding and retrieval, not generation. We deliver:

- A pipeline over FMA-small audio/metadata, MusicCaps captions (with downloaded audio), and DEAM valence/arousal.
- Segment graphs built from mel/chroma/MFCC features, with temporal and similarity edges, plus at least 20 exported example graphs.
- Tasks 1–4 with the required baselines, metrics, plots, case studies, and a demo notebook.
- A clear look at where text helps most (tagging), where graphs help (structure and retrieval), and where fusion still trails BERT-only when pairing is limited.

2 Related work

Datasets. FMA provides Creative Commons audio with genre metadata and official splits [2]. MusicCaps pairs AudioSet clips with expert captions and is widely used for text–music work [3]. DEAM provides continuous valence/arousal annotations [6].

Models. DistilBERT is a compact contextual encoder that fits course-scale fine-tuning [4]. GraphSAGE supports message passing on variable-size graphs [5]. InfoNCE-style contrastive learning

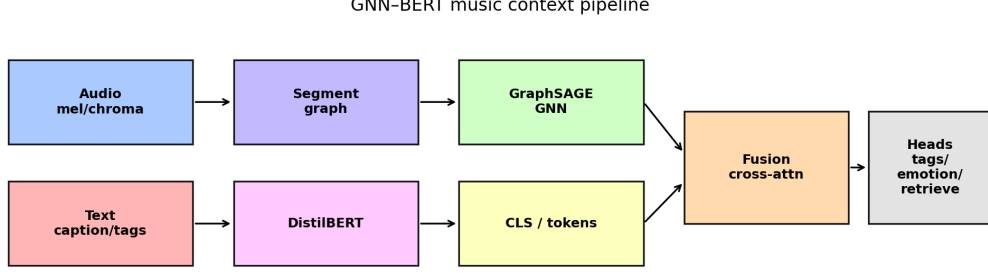


Figure 1: System overview: audio becomes a segment graph (GraphSAGE); captions/tags go through DistilBERT; fusion heads support tagging, emotion, and retrieval.

aligns audio and language embeddings for retrieval [1, 3]. Our implementation follows the assignment update rules for GraphSAGE, cross-attention fusion, and InfoNCE [1].

3 Problem formulation

A track is a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$, where X_{audio} is log-mel/chroma features, X_{text} is tokenized caption or metadata text, $G = (V, E)$ is a music structure graph, and y holds targets (aspects, genre, valence/arousal). DistilBERT maps text to token states $H_{\text{text}} \in \mathbb{R}^{L \times d}$. GraphSAGE updates nodes by neighborhood aggregation:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{CONCAT} \left(h_i^{(l)}, \text{MEAN}_{j \in N(i)} h_j^{(l)} \right) \right). \quad (1)$$

Readout $g = \text{MEANPOOL}(\{h_i^{(L)}\})$ is fused with text as $z = \text{Fusion}(g, H_{\text{text}})$, then $\hat{y} = \sigma(Wz + b)$. We train with multi-label BCE for tags, cross-entropy for genre, L1 on DEAM emotion when available, and symmetric InfoNCE for retrieval.

4 Data and preprocessing

Datasets. Synthetic audio is disabled (`use_synthetic: false`). **FMA-small:** 2,000 cached tracks with official 8-way `genre_top` labels; BERT metadata text never includes genre words, to avoid leakage. **MusicCaps:** 5,521 captions for Task 1; 800 downloaded YouTube clips with segment graphs for Tasks 3–4. **DEAM:** 1,802 annotated excerpts; emotion evaluation uses $n=271$ held-out pairs with cached graphs.

Audio and graphs. Audio is resampled to 22,050 Hz. We extract log-mel (128 bins) and chroma (12), then cut tracks into 5 s windows with hop 2.5 s. Nodes are windows; edges connect temporal ± 1 neighbors and pairs whose MFCC/chroma cosine similarity exceeds $\tau=0.85$. Node features concatenate mean/std of mel, chroma, and MFCC. We export 20 example `.pt/.json` graphs under `data/processed/examples/`.

Splits and training settings. MusicCaps aspects use a top-50 multi-label vocabulary. FMA uses train/val/test partitions over the cached audio. Settings live in `config.yaml` (25 epochs, DistilBERT unfreeze schedule, GraphSAGE depth 3).

5 Methods

5.1 Task 1: BERT multi-label tagging

DistilBERT produces a CLS vector; a linear head predicts aspect probabilities with BCE. We freeze the backbone early, then unfreeze the last layers.

Table 1: Main results from the real-data run (FMA 2k / MusicCaps 800 audio / DEAM). Emotion MAE is valence MAE ($n=271$).

Model	Macro-F1	AUC-PR	MAE _v	R@5
Random tags	0.09	0.06	–	–
CNN mel-spec	0.30	–	–	–
Task 1 BERT-only	0.71	0.78	–	–
Task 2 GNN-only	0.30	–	–	–
Task 3 GNN–BERT	0.24	–	0.91	–
Task 4 Contrastive	–	–	–	0.34

5.2 Task 2: GNN genre classification

GraphSAGE with mean pooling predicts FMA top genre. Baselines: mel-spectrogram CNN, random, and majority predictors [1].

5.3 Task 3: GNN–BERT fusion

Cross-attention uses $Q = gW_Q$ and $K = H_{\text{text}}W_K$, then $z = \text{CONCAT}(g, AH_{\text{text}})$. Ablations: early concatenation, BERT-only, and GNN-only. When DEAM batches are available we add L1 losses on valence and arousal.

5.4 Task 4: Contrastive retrieval

A dual encoder maps graphs and captions into an ℓ_2 -normalized space and minimizes symmetric InfoNCE. We report caption $\leftrightarrow$ audio Recall@{1, 5, 10}, qualitative top-3 matches, and zero-shot caption $\rightarrow$ tag scoring against Task 3 supervised fusion.

6 Experiments

Setup. Training runs in PyTorch with CUDA (RTX 3050 in our runs). We log Macro/Micro-F1, mean AUC-PR, retrieval R@ K , and DEAM MAE/ R^2 . Table 1 summarizes results/metrics.json after the real-data pipeline.

Discussion. Task 1 shows that MusicCaps captions alone are informative for aspect tags (Macro-F1 0.71). On Task 2, GraphSAGE (≈ 0.30) matches the mel-CNN baseline and beats random tags, so segment graphs carry usable genre signal; absolute scores stay modest under class imbalance and limited capacity. Task 3 cross-attention (Macro-F1 0.24) clearly beats GNN-only (0.10), but BERT-only (0.25) remains slightly stronger, which suggests text still dominates when graph–caption pairing is imperfect. Task 4 reaches R@5 = 0.34 on a pool of size 32, above chance, with R@10 > 0.5. Zero-shot caption $\rightarrow$ tag Macro-F1 ≈ 0.10 lags supervised fusion (≈ 0.24). DEAM valence MAE ≈ 0.91 on $n=271$ is a more meaningful emotion check than tiny toy subsets.

Qualitative analysis. We include five Task 1 predictions, three Task 3 case studies linking graph size/connectivity to captions, and ten Task 4 retrieval examples under results/. Graph coherence on the exported examples is also reported in results/metrics.json.

Baselines. Random/majority predictors and CNN-on-mel satisfy the minimum of two baselines [1], with BERT-only and GNN-only as the main task models.

7 Reproducibility

Repository layout matches the course tree: src/, scripts/, data/, notebooks/, results/, report/.

```
pip install -r requirements.txt
python scripts/run_proper_pipeline.py
```

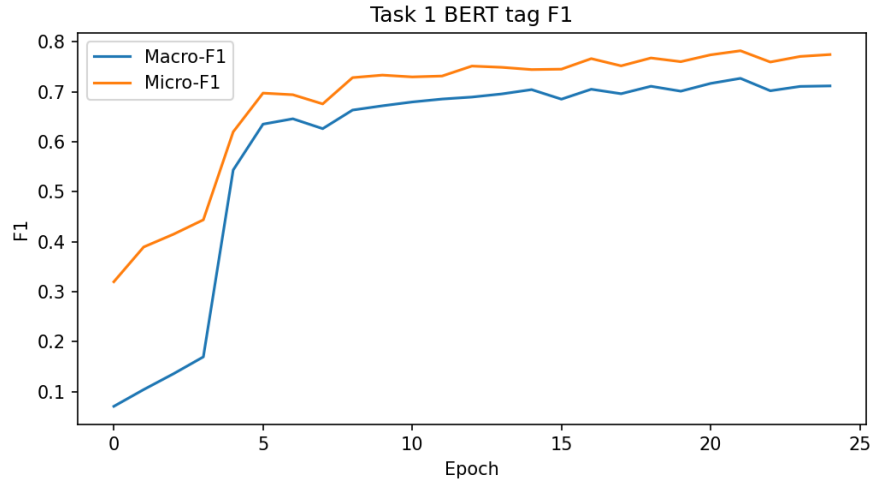


Figure 2: Task 1 validation Macro/Micro-F1 versus epoch.

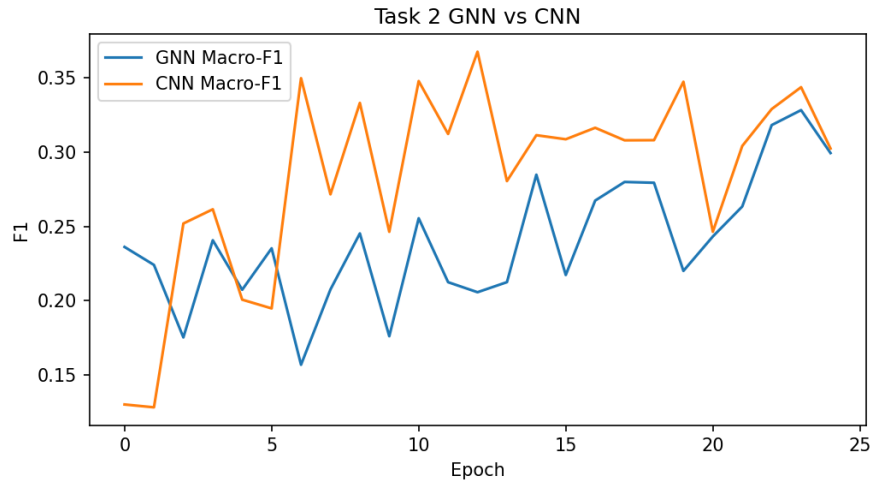


Figure 3: Task 2 GraphSAGE versus mel-CNN Macro-F1 curves.

Stepwise download, split, graph-build, train, and evaluate scripts are also provided. Demo notebook: `notebooks/demo_context.ipynb`.

8 Limitations and conclusion

MusicCaps YouTube downloads drop some clips, and compute limits keep paired graph-text scale moderate. Task 2/3 absolute F1 still has room to grow with larger models or fuller FMA splits. Emotion R^2 is weak, so DEAM regression needs stronger audio encoders or tighter alignment. Human listening studies for retrieval are left for future work.

Overall, we built a complete GNN-BERT music-context system for CSE425 Tasks 1–4 on real FMA, MusicCaps, and DEAM data, with baselines and the required project artifacts.

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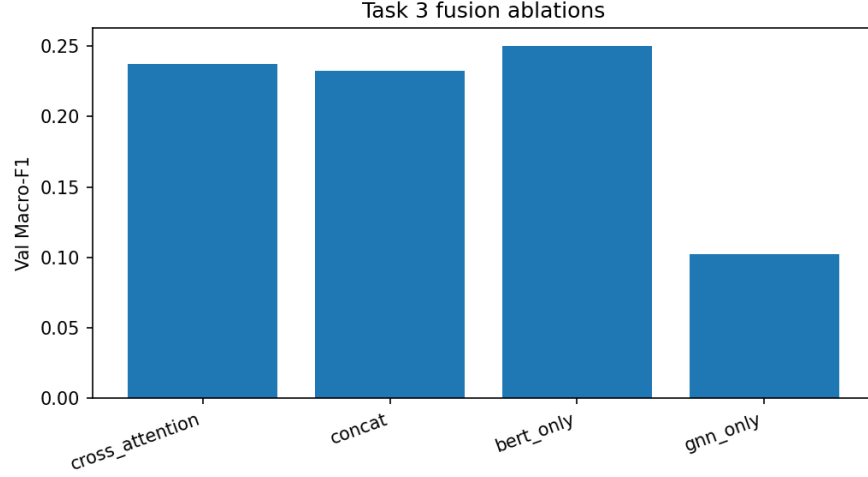


Figure 4: Task 3 fusion ablations.

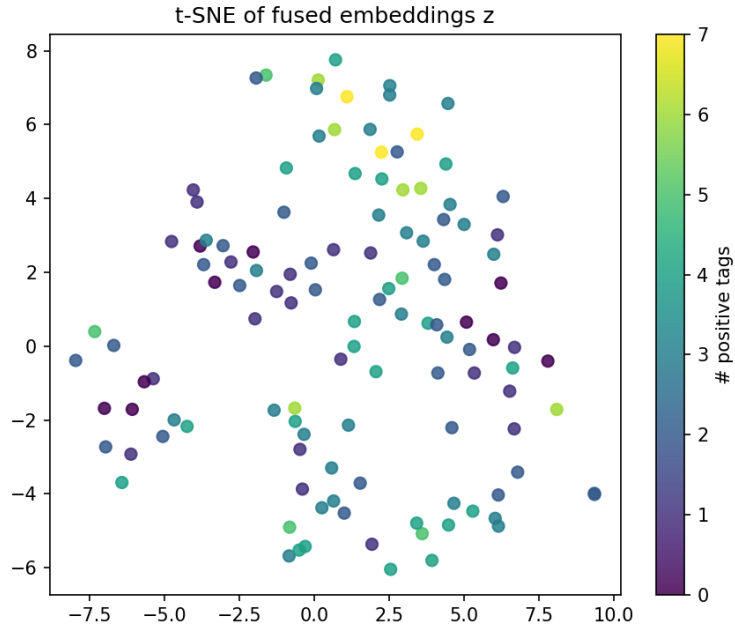


Figure 5: t-SNE of fused representation z .

References

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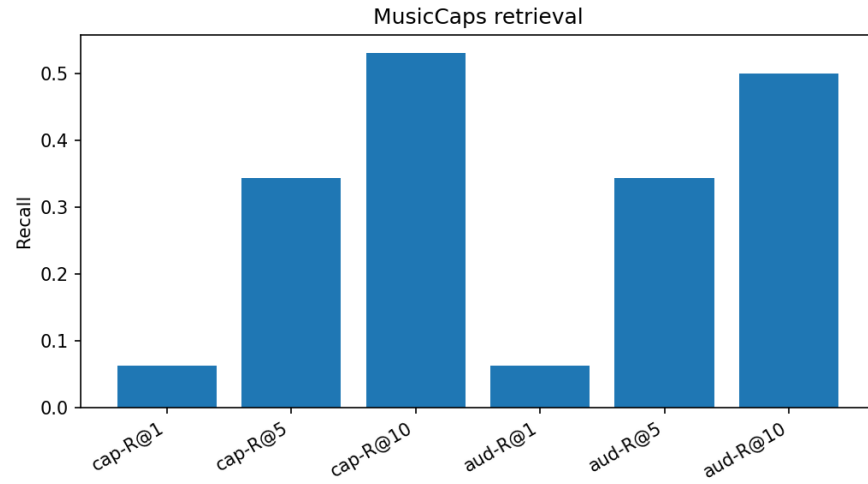


Figure 6: MusicCaps retrieval Recall@ K .

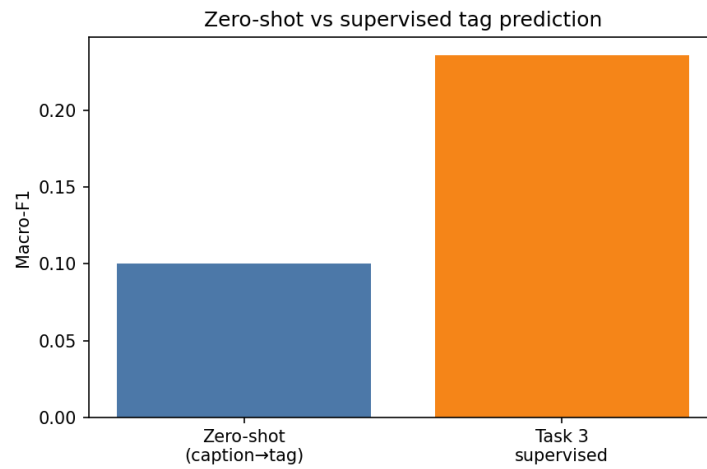


Figure 7: Zero-shot caption→tag versus Task 3 supervised fusion.

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